

AMHS Automated Cleanroom FOUP Stocker

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-STK-001

NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

1. Machine Overview & Manufacturing Context

Equipment Model:	AMHS Automated Cleanroom FOUP Stocker	Document ID:	VAI-MAN-STK-001
Machine Type Key:	stocker	Cleanroom Bay / Stage:	Bay 1: Wafer Dicing & Prep
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Automated Material Handling System (AMHS) cleanroom stocker for storing, buffering, and N2-purging 300mm Front Opening Unified Pods (FOUPs) between wafer prep and assembly bays.

Process Flow: Overhead Hoist Transport (OHT) vehicles deposit FOUP carriers at load ports. A high-speed dual-axis Cartesian robotic crane transfers cassettes into storage bays while ultra-pure N2 purge nozzles prevent surface oxidation and moisture adsorption.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
Robotic Crane Gripper	Transfers 300mm FOUPs between load ports and storage shelving.	Crane velocity (m/s), vibration (mm/s), gripper clamping force (N).	Crane track chatter (>0.3 mm/s), gripper timing delays.
N2 Purge Manifold	Delivers continuous 99.999% purity N2 gas into FOUP purge ports.	N2 purge pressure (kPa), mass flow rate (SLPM), moisture (ppb).	Purge pressure drop (<40 kPa), manifold solenoid valve leakage.
Load Port Docking Mechanism	Clamps FOUP and opens automated door under ISO 1 mini-environment.	Door seal vacuum level (kPa), latch actuation torque (Nm).	Door seal vacuum drop, latch alignment timeout.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
Robotic Crane Vibration vibration_crane	mm/s	0.05 – 0.25	0.25 – 0.6	0.6 – 1.5	Higher is Worse
N2 Purge Pressure pressure_n2	kPa	45 – 60	30 – 45	0 – 30	Lower is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 4,500 operating hours. Model: $RUL = BaseLife * (1 - (0.55 * wear_crane_vib + 0.45 * wear_n2_press))$

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
Crane Vibration	vibration_crane	mm/s	0.55	0.15	0.8	HIGHER IS WORSE
N2 Purge Pressure	pressure_n2	kPa	0.45	52	25	LOWER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Crane Positioning Timeout / Jitter SYM-STK-001	HIGH	vibration_crane	• Linear encoder glass scale dirt • Drive belt tension loss • Guide roller flat spot	Clean optical encoder scale with isopropanol and adjust timing belt tension.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: Crane Guide Rail Roller Wear & Micro-Chatter (SCEN-STK-001)	Severity: WARNING
Telemetry Sensor Pattern:	vibration_crane > 0.45 mm/s
Possible Root Causes:	Urethane drive roller wear Rail alignment deviation
Recommended Corrective Action:	Replace crane guide roller assemblies and execute dynamic axis calibration.
Verification & Recovery Steps:	Inspect roller surface > Run 10-cycle automated pickup repeatability test (< 0.1mm tolerance)